

SALESTORM – Functional and Non-Functional Requirements

1. Functional Requirements

FR1 – User Authentication

The system shall allow customers to register, log in, and access authorized features.

FR2 – Product Browsing

The system shall allow customers to browse products and view product details, price, and availability.

FR3 – Cart Management

The system shall allow customers to add products to the cart, update quantities, and remove products.

FR4 – Inventory Management

The system shall maintain product inventory and track available, reserved, and sold quantities.

FR5 – Inventory Reservation

The system shall temporarily reserve inventory when a customer initiates a purchase.

FR6 – Concurrent Purchase Handling

The system shall handle multiple customers attempting to purchase the same product simultaneously without overselling.

FR7 – Reservation Expiry

The system shall automatically release inventory when a reservation expires, payment fails, or the checkout process is abandoned.

FR8 – Idempotent Purchase Requests

The system shall prevent duplicate reservations and duplicate orders when the same request is submitted multiple times.

FR9 – Checkout

The system shall validate the customer's purchase information and initiate the payment process.

FR10 – Payment Processing

The system shall handle payment success, payment failure, timeout, retry, and duplicate payment scenarios.

FR11 – Order Creation

The system shall create an order after successful payment and maintain the complete order lifecycle.

FR12 – Order Status Management

The system shall support order states such as CREATED, PAYMENT_PENDING, CONFIRMED, PROCESSING, SHIPPED, OUT_FOR_DELIVERY, and DELIVERED.

FR13 – Failure Recovery

The system shall recover from scenarios such as payment success followed by temporary Order Service failure.

FR14 – Shipment Management

The system shall send confirmed orders for shipment and maintain shipment status.

FR15 – Notification

The system shall notify customers about relevant payment, order, shipment, and delivery events.

2. Non-Functional Requirements

NFR1 – Scalability

The system shall support approximately 10,000 requests per second under normal traffic and be designed to scale up to approximately 500,000 requests per second during flash-sale traffic.

NFR2 – High Concurrency

The system shall support thousands of simultaneous purchase requests for the same product without system failure.

NFR3 – Inventory Consistency

The system shall guarantee that successful sales never exceed the available inventory.

Example: If only 100 units are available, at most 100 successful purchases shall be allowed.

NFR4 – Availability

The system shall remain available and resilient even when individual services or external dependencies temporarily fail.

NFR5 – Reliability

The system shall provide retries, timeouts, idempotency, reconciliation, and recovery mechanisms for critical operations.

NFR6 – Performance

The system shall provide low-latency responses during normal and flash-sale traffic while controlling database and inventory contention.

NFR7 – Fault Tolerance

The system shall gracefully handle failures in payment gateways, databases, message queues, and downstream services.

NFR8 – Security

The system shall provide authentication, authorization, HTTPS, rate limiting, input validation, secure payment-data handling, audit logging, and secrets management.

NFR9 – Observability

The system shall provide metrics, structured logging, distributed tracing, monitoring, and alerts for critical operations.

NFR10 – Maintainability

The architecture shall have clearly separated services and responsibilities so that new payment providers, pricing strategies, or delivery partners can be added with minimal impact.

NFR11 – Recoverability

The system shall recover safely from partial failures and ensure that inventory, payment, and order states eventually become consistent.

NFR12 – Auditability

Critical inventory, payment, reservation, and order operations shall be traceable through logs and transaction/event records.